

Linear Simultaneous Equations - Solutions

LET – Mathematics and Statistics

Two Unknowns

Solve the following systems of equations.

1.

$$5x + 2y = 24$$

$$y = x + 5$$

Solution:

This is best solved by substitution. Substitute the expression for y from the second equation into the first:

$$5x + 2(x + 5) = 24$$

$$5x + 2x + 10 = 24$$

$$7x = 14$$

$$x = 2$$

Now substitute $x = 2$ back into $y = x + 5$:

$$y = 2 + 5$$

$$y = 7$$

Answer: $x = 2, y = 7$

2.

$$8x - y = 34$$

$$3x + y = 10$$

Solution:

This is best solved by elimination. Add the two equations to eliminate y :

$$(8x - y) + (3x + y) = 34 + 10$$

$$11x = 44$$

$$x = 4$$

Now substitute $x = 4$ back into the second equation:

$$3(4) + y = 10$$

$$12 + y = 10$$

$$y = -2$$

Answer: $x = 4, y = -2$

3.

$$10x + 3y = 25$$

$$2y - 6x = 8$$

Solution:

First, rearrange the equations to align the variables and label them:

$$10x + 3y = 25 \quad (1)$$

$$-6x + 2y = 8 \quad (2)$$

Use elimination. Multiply (1) by 2 and (2) by 3 to make the y coefficients equal:

$$20x + 6y = 50 \quad (3)$$

$$-18x + 6y = 24 \quad (4)$$

Subtract equation (4) from (3):

$$(20x - (-18x)) + (6y - 6y) = 50 - 24$$

$$38x = 26$$

$$x = \frac{26}{38} = \frac{13}{19}$$

Substitute $x = 13/19$ into equation (2):

$$-6\left(\frac{13}{19}\right) + 2y = 8$$

$$-\frac{78}{19} + 2y = 8$$

$$2y = 8 + \frac{78}{19} = \frac{152 + 78}{19} = \frac{230}{19}$$

$$y = \frac{115}{19}$$

Answer: $x = \frac{13}{19}, y = \frac{115}{19}$

4.

$$P = \frac{71}{2} - 5Q$$

$$7P - 3Q = \frac{193}{2}$$

Solution:

Use substitution. Substitute the expression for P into the second

equation:

$$\begin{aligned}7\left(\frac{71}{2} - 5Q\right) - 3Q &= \frac{193}{2} \\ \frac{497}{2} - 35Q - 3Q &= \frac{193}{2} \\ \frac{497}{2} - 38Q &= \frac{193}{2} \\ \frac{497 - 193}{2} &= 38Q \\ \frac{304}{2} &= 38Q \\ 152 &= 38Q \\ Q &= 4\end{aligned}$$

Substitute $Q = 4$ back into the first equation:

$$\begin{aligned}P &= \frac{71}{2} - 5(4) \\ P &= \frac{71}{2} - 20 = \frac{71 - 40}{2} = \frac{31}{2}\end{aligned}$$

Answer: $P = \frac{31}{2}, Q = 4$

Three Unknowns

5.

$$y = 50 - 5x$$

$$z = 4x + 7$$

$$y = x + 38$$

Solution:

We have two expressions for y . Set them equal to each other:

$$50 - 5x = x + 38$$

$$12 = 6x$$

$$x = 2$$

Substitute $x = 2$ into the other equations to find y and z :

$$y = 2 + 38 = 40$$

$$z = 4(2) + 7 = 15$$

Answer: $x = 2, y = 40, z = 15$

6.

$$x - y + 4z = 2$$

$$2x + y + z = 15$$

$$5x + z = 21$$

Solution:

The third equation already lacks a y . Add the first two equations to

eliminate y :

$$\begin{aligned}(x - y + 4z) + (2x + y + z) &= 2 + 15 \\ 3x + 5z &= 17 \quad (4)\end{aligned}$$

Now solve the system formed by equations (3) and (4):

$$\begin{aligned}5x + z &= 21 \quad (3) \\ 3x + 5z &= 17 \quad (4)\end{aligned}$$

From (3), $z = 21 - 5x$. Substitute this into (4):

$$\begin{aligned}3x + 5(21 - 5x) &= 17 \\ 3x + 105 - 25x &= 17 \\ -22x &= -88 \\ x &= 4\end{aligned}$$

Find z using $z = 21 - 5x$, then find y using equation (2):

$$\begin{aligned}z &= 21 - 5(4) = 1 \\ 2(4) + y + (1) &= 15 \implies 9 + y = 15 \implies y = 6\end{aligned}$$

Answer: $x = 4, y = 6, z = 1$

7.

$$\begin{aligned}3x + 2y - 3z &= 2 \\ 2x + y + z &= 10 \\ x + 3y + 2z &= 12\end{aligned}$$

Solution:

Label the equations (1), (2), (3). First, eliminate y using (1) and (2).

Multiply (2) by -2 and add to (1):

$$\begin{aligned}(3x + 2y - 3z) - 2(2x + y + z) &= 2 - 2(10) \\ (3x + 2y - 3z) + (-4x - 2y - 2z) &= 2 - 20 \\ -x - 5z &= -18 \implies x + 5z = 18 \quad (4)\end{aligned}$$

Next, eliminate y using (2) and (3). Multiply (2) by -3 and add to (3):

$$\begin{aligned}(x + 3y + 2z) - 3(2x + y + z) &= 12 - 3(10) \\ (x + 3y + 2z) + (-6x - 3y - 3z) &= 12 - 30 \\ -5x - z &= -18 \implies 5x + z = 18 \quad (5)\end{aligned}$$

Solve the system of (4) and (5). From (5), $z = 18 - 5x$. Substitute into (4):

$$\begin{aligned}x + 5(18 - 5x) &= 18 \\ x + 90 - 25x &= 18 \\ -24x &= -72 \\ x &= 3\end{aligned}$$

Find z using $z = 18 - 5x$, then find y using equation (2):

$$\begin{aligned}z &= 18 - 5(3) = 3 \\ 2(3) + y + (3) &= 10 \implies 9 + y = 10 \implies y = 1\end{aligned}$$

Answer: $x = 3, y = 1, z = 3$